Computationally Invisible

Charting the NLP Landscape of Iranic Languages

Hojjat Mokhtarabadi[1*] · Amir Hossein Kargaran[2*] · Ali Shakeri[1] · Saba Shafiei[1] · Nafiseh Nikeghbal[3] · Mohammad Hossein Manshaei[4]

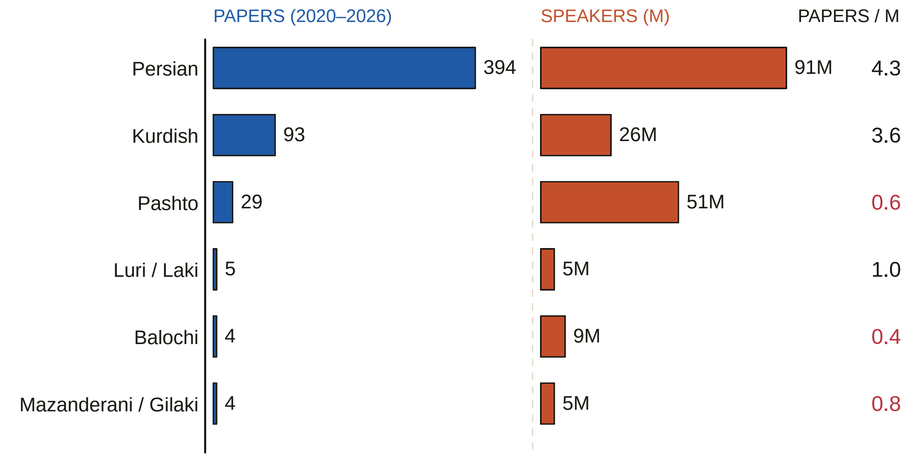
[1]Isfahan University of Technology [2]LMU Munich [3]TU Munich [4]Hunter College and Graduate Center, City University of New York [*]Equal contribution

SURVEY · 495 PAPERS · 2020–2026

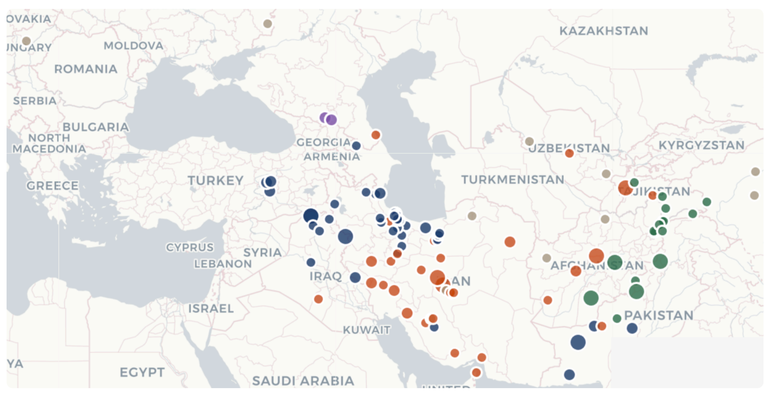
80% of surveyed Iranic NLP papers (394 of 495) address one of 101 varieties: Iranian Persian.

Pashto (**51M speakers**) is covered by 29 papers and Balochi (**9M**) by 4. For most other varieties, no dedicated paper, benchmark or tool was found.

Paper count and speaker population



495 papers, 2020 – May 2026. Papers covering several languages count once per language. Red: fewer than one paper per million speakers.



Where Iranic languages are spoken

- Southwestern
- Northwestern
- Eastern
- Ossetic
- Historical

All 101 ISO 639-3 coded varieties at Glottolog centroid coordinates; dot size reflects speaker population. From the Caucasus to Xinjiang and from the North Caucasus to Oman.

Scan for the companion website
language maps · resource browser · live paper index
sigiranic.github.io

THE SURVEY

101
ISO 639-3 coded varieties in 4 living branches

200M+
speakers, Caucasus to Xinjiang, Russia to Oman

5,878 → 495
candidates filtered and annotated with Qwen3.5-9B, every label checked by two authors

01 · THE FAMILY

Major Iranic languages by branch: L1 speakers and NLP resource level

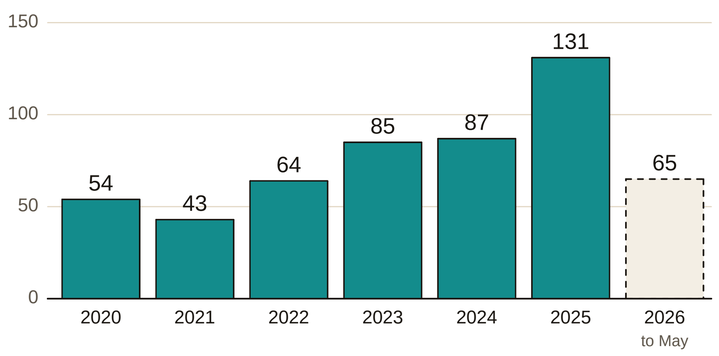
Iranic · 101 ISO 639-3 codes

SOUTHWESTERN			NORTHWESTERN			EASTERN			OSSETIC		
Persian	91M	HIGH	Kurdish	26M	LOW	Pashto	51M	LOW	Ossetic	0.6M	VERY LOW
Dari	15M	MED	Balochi	9M	VERY LOW	Shughni	0.1M	VERY LOW			
Tajik	8M	LOW	Gilaki	3M	VERY LOW	Wakhi	0.06M	VERY LOW			
Luri / Bakhtiari	5M	VERY LOW	S. Kurdish	3M	VERY LOW	Yaghnobi	0.01M	VERY LOW			
Hazaragi	3M	VERY LOW	Zazaki	2–4M	VERY LOW						
Laki	0.7M	VERY LOW	Mazanderani	1.4M	VERY LOW						
			Talysh	0.8M	VERY LOW						
			Gorani	0.4M	VERY LOW						

L1 speakers · NLP resource level: high medium low very low

02 · PUBLICATIONS

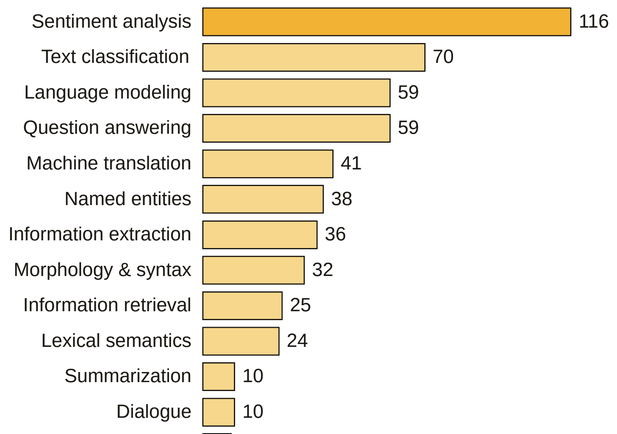
Publications per year, 2020–2026



Kurdish and Pashto grow modestly; other varieties stay near zero. 2026 is partial (search cut-off May 2026).

03 · TASKS

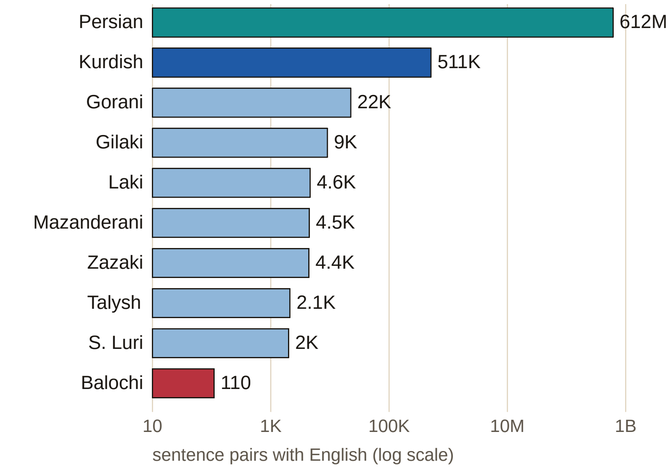
Publications by NLP task



Papers spanning several tasks count once per task.

04 · PARALLEL DATA

Parallel sentence pairs with English



Between regional varieties: a few thousand pairs at most. Transfer runs through Persian or English.

05 · OPEN CHALLENGES

Open challenges and research priorities

Resource concentration
Persian ≈ 80% of papers; Pashto (51M) has fewer resources than Kurdish.
Priority: shared tasks and central data hubs

Script heterogeneity
Perso-Arabic, Latin, Cyrillic; inconsistent ya/kaf code points; ZWNJ bugs.
Priority: a unified Iranic normalization library

Dialectal fragmentation
Most benchmarks evaluate only standard written forms.
Priority: colloquial and regional benchmarks

Cultural misalignment and saturation
Persian sentiment, QA, NER near saturation; many benchmarks translated from English.
Priority: benchmarks co-created with native speakers

Endangered varieties
Pamir and Central Plateau varieties may disappear within decades.
Priority: multilingual speech models, community recording

Structural inequity
Sanctions limit compute, API access and travel for many researchers.
Priority: compute and access programs

IMMEDIATE PRIORITIES

Raw text
Iranic text on the open web, with correct language tags.

Translation benchmarks
FLORES+ still covers the same 6 Iranic varieties as FLORES-200.

Evaluation beyond translation
At least one natively authored benchmark per variety.

Code and prompts: github.com/sigiranic/sigiranic.github.io

Companion website: sigiranic.github.io